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Interface Circuits for a Six-Phase Variable Reluctance Energy Harvester

A comparative study of rectification and power conditioning
topologies

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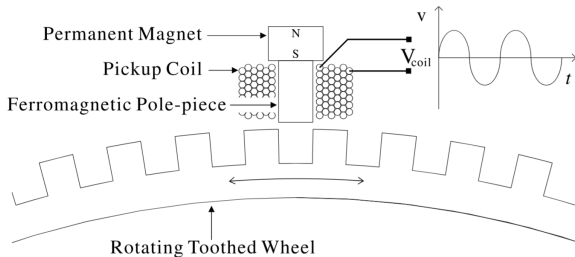
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Why energy harvesting?



- WSN and IoT nodes are everywhere, **collecting data** for many different purposes.
- Batteries are costly to maintain, have a finite life and are often **impossible to replace**, and they are bad for the environment.
- With **energy harvesting** the node powers itself from the energy already available around it.

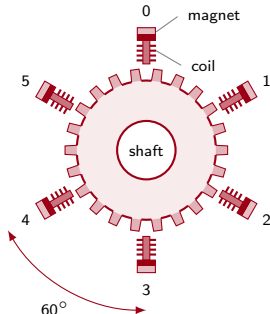


The six-phase VREH



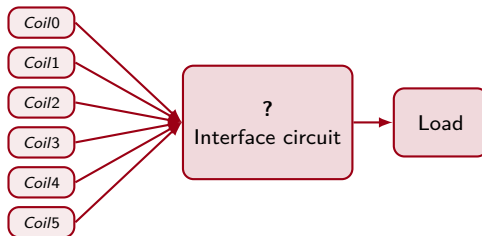
Six pickup units around the shaft, electrically 60° apart, evolved from railway prototypes into a design for large vehicles.

Target range **100 - 400 rpm** (20 - 80 km/h), with a minimum phase voltage of 1.1 V peak to peak.



Each unit is a permanent magnet over its pickup coil.

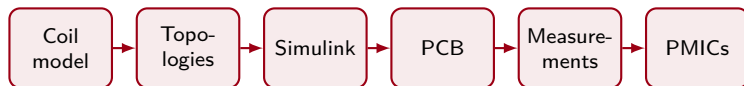
six coils, 60° apart



- Six phases give more power, but also **six time-shifted AC signals** that all have to be rectified.
- Earlier work only compared rectifiers for a **single phase**.

Research question

Which interface circuit extracts the most power, over the whole speed range?



From theory to a real prototype, with **every topology measured on the same board** and on the same test bench.

Coil model and matching

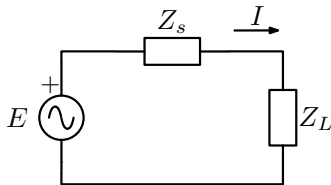
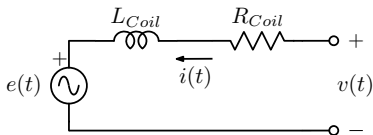


- Each coil is a source in series with R_{Coil} and L_{Coil} , delayed by 60° from the previous one ($k = 0, 1, \dots, 5$),

$$e_k(t) = E_{pk} \sin\left(\omega_e t - k \frac{\pi}{3}\right)$$

- At low speed a **resistive** load is enough.
- At higher speed $\omega_e L_{\text{Coil}}$ breaks the matching.
- A series capacitor C_m cancels it, one frequency at a time,

$$C_m = \frac{1}{\omega_e^2 L_{\text{Coil}}}$$



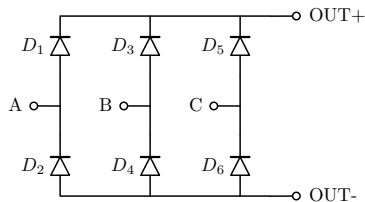
Strategy 1 - FWR bridges



Two three-phase groups ($k = 0, 2, 4$ and $k = 1, 3, 5$), each on a standard bridge.

- The **Star (Y)** gives a line voltage $\sqrt{3}$ times the phase EMF, so the two-diode drop costs less.
- The **Delta (Δ)** gives the coil EMF itself, with no gain but no floating node.

The Star should give the **higher open-circuit voltage**.



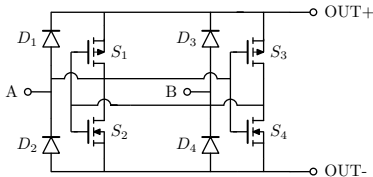
Two diodes always in the current path.

Three two-phase groups, pairing phases 180° apart in **anti-series**

$$v_A - v_B = e_k - e_{k+3} = 2 e_k$$

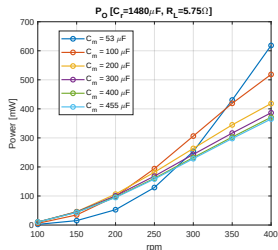
ideally double the voltage.

- Four MOSFETs with **cross-coupled gates**, so no gate driver is needed.
- They bypass the diodes, trading the fixed drop for a small R_{on} .

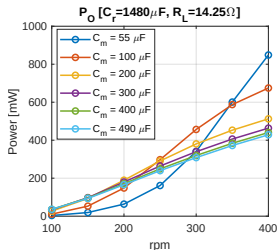


NVC for one anti-series pair.

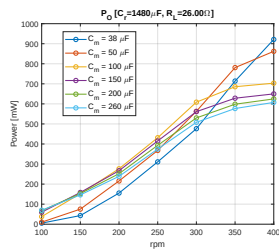
Simulation results



Delta



Star



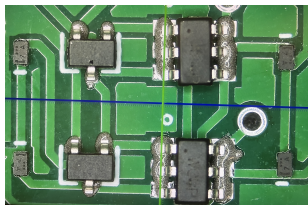
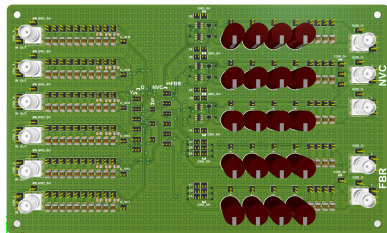
NVC

- Power against speed, for several fixed C_m .
- At 100 rpm, the worst case, the NVC gives **69.0 mW**, about $6\times$ the Delta (10.7) and nearly double the Star (36.7).
- The NVC is also the most efficient at every speed.

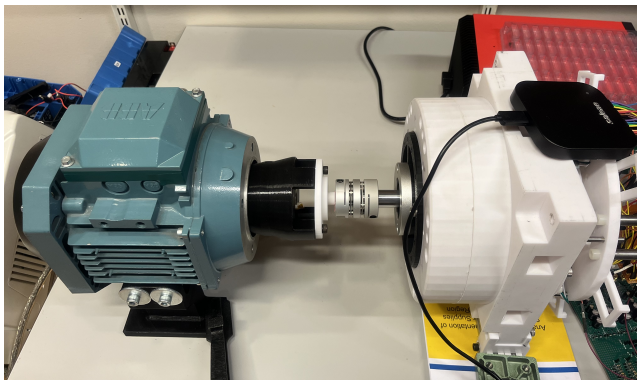
The prototype board



- Custom PCB, Altium Designer.
- Delta, Star and NVC selected by moving jumpers, plus a jumper bank for C_m .
- Hand assembled with soldering iron and hot plate.

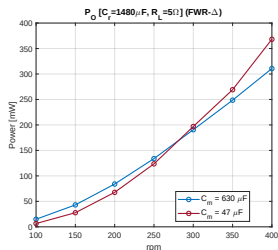


One NVC bridge during assembly.

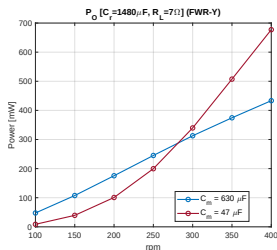


Motor coupled to the VREH.

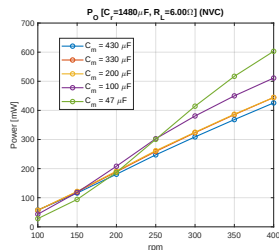
An inverter drives the motor and sets the speed precisely over the whole 100-400 rpm range.



Delta



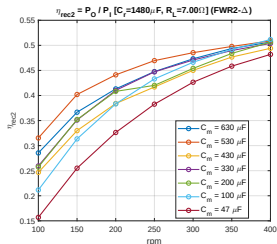
Star



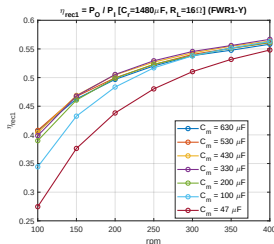
NVC

- At 100 rpm the NVC still wins with 57.7 mW, against 14.9 (Delta) and 47.6 (Star).
- Above 250 rpm the **Star overtakes the NVC**, 433.3 against 425.7 mW at 400 rpm.
- Simulation did not predict this crossover.

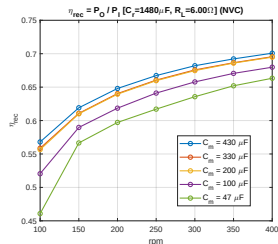
Measured efficiency



Delta



Star



NVC

- Best efficiency at every speed for the NVC, up to $\eta_{Rec} \approx 0.70$.
- Delta stops at ≈ 0.48 - 0.51 , the weakest as expected, and Star reaches ≈ 0.55 - 0.57 .

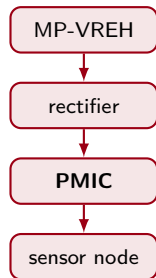
Measured power differs from simulation by 8% for the Star, up to almost 30% for the NVC.

	Delta	Star	NVC
Mean deviation of P_O from simulation	16%	8%	30%

Why?

- The ceramic C_m lose capacitance under the real voltage, which grows with speed.
- Component tolerances and the parasitic resistance of the bridges.
- Sources that are not perfectly sinusoidal.

PMIC	V_{in} (V)	Star up to
DS-AEM13920_VR	0.120-4.6	250 rpm
DS-AEM10941_VR	0.05-5	250 rpm
bq25570	0.1-5.1	250 rpm
SPV1050 (Boost)	0.15-5.3	250 rpm
SPV1050 (Buck-Boost)	0.15-18	full range
SPV1040	0.3-5.5	300 rpm



The PMIC turns the rectified power into a stable supply.

- Delta and NVC stay below 4 V and are **compatible with every chip tested**.
- The Star reaches 7 V and needs a wide-input converter or a clamp.

Conclusions

- **NVC and Star** are the two strongest candidates, one per end of the speed range; Delta is always the weakest.
- NVC and Delta fit any PMIC tested; the Star needs protection at high speed, its open-circuit voltage can damage the input.

Limitation

The harvester prototype is 3D-printed and sensitive to vibration at high speed, which makes the measurements harder to repeat.

Future work

Adaptive compensation tracking C_m with speed, and a full hardware test up to a real PMIC, to measure the efficiency from harvester to load.

Thank you for your attention!